Tritium

- Production
  - Produced in the upper atmosphere by interaction of nitrogen, and to a lesser extent, oxygen with fast neutrons from cosmic rays.

$$^{14}_{7}N + n \rightarrow ^{12}_{6}C + ^{3}_{1}H$$

- Inventory
  - 1 TU = [T]/[H] = 1×10^{-18} TU=Tritium Unit
  - 1 TU = 3.2 pCi/kg H_2O = 0.12 Bq/kg H_2O
  - Natural level in precipitation ~ 5 TU
  - 'Bomb tritium' max. levels ~1000 to 2000 TU
- In 1973
  - natural T: ~2.5%
  - nuclear power plant T: 0.02%
  - bombs: 97.5%

JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 97, NO. C4, PAGES 5481–5492, APRIL 15, 1992

A Model Function of the Global Bomb Tritium Distribution in Precipitation, 1960–1986

SCOTT C. DONEY,[1] DAVID M. GLOVER, AND WILLIAM J. JENKINS
Chemistry Deptartment, Woods Hole Oceanographic Institution, Woods Hole, Massachusetts

Note change in scale between N.H. and S.H.

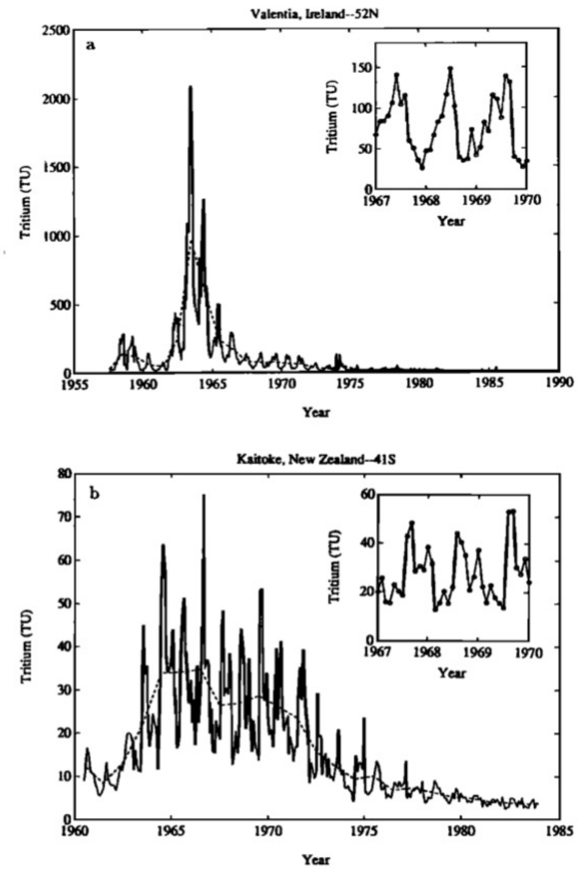
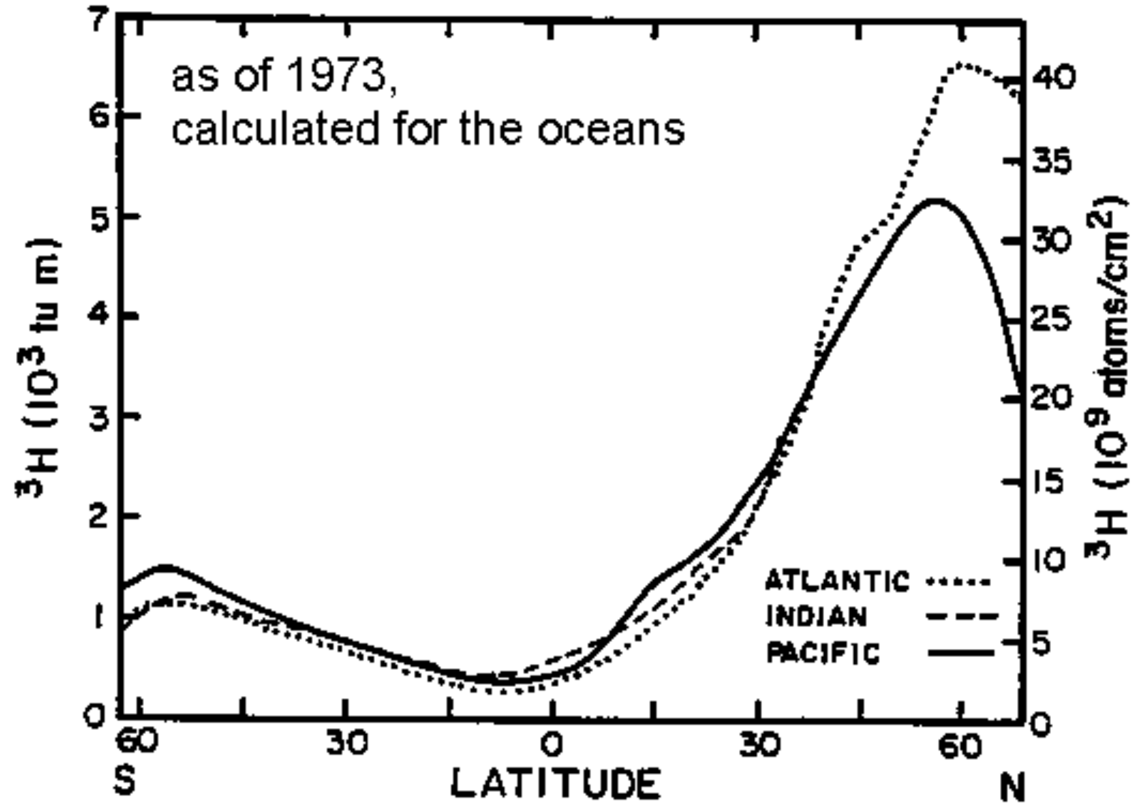


Fig. 1. Monthly tritium concentrations in precipitation plotted versus time from (*a*) Valentia, Ireland (51.9°N), and (*b*) Kaitoke, New Zealand (41.1°S). The dashed lines connect the annual average tritium concentrations at each station. Note the change of scale between the northern and southern hemisphere stations. The insets highlight the seasonal tritium cycle for the years 1967–1970.

N.H.

S.H.



- ^3H produced by atmospheric nuclear weapons tests (late 1950s and early 1960s).
- Oxidized to water and rained out over the oceans and continents over the years after the tests.
- Since the tests were largely in the Northern Hemisphere, most of the ^3H (about 80%) has been deposited north of the equator, predominantly at higher latitudes.

Tritium

- Tritium is the radioactive isotope of hydrogen and has a natural component of ca. 5 TU
- ‘Bomb’ Tritium is a transient tracer released mainly during the surface nuclear weapons tests in the 1950’s and early 1960’s and masks the natural background more or less completely.
- Most of the tritium can be found in the northern hemisphere due to its release pattern
- The ocean is the main sink for tritium in the environment.
- The tritium concentrations in the environment are approaching natural levels since ca. 5 decades have been passed since the major releases during the surface nuclear weapons tests

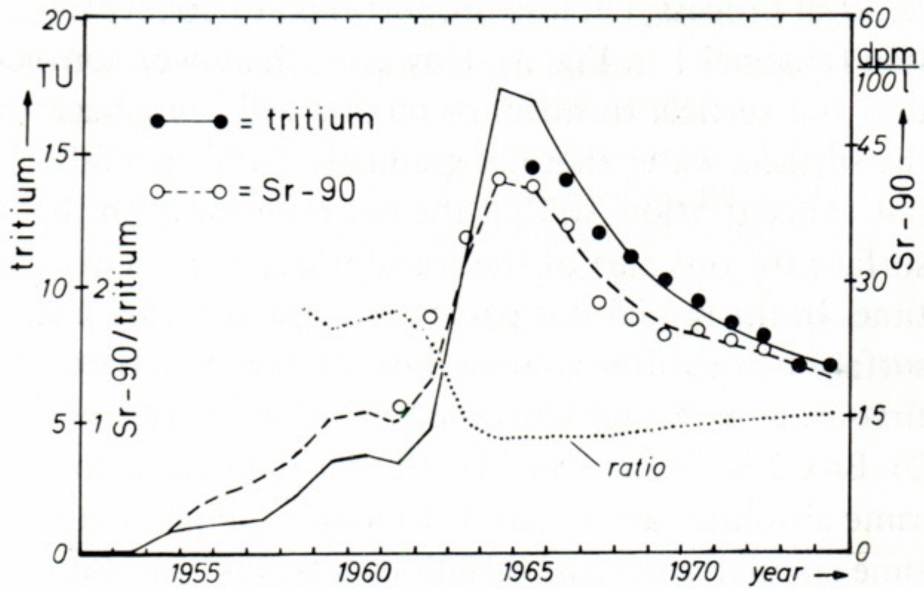
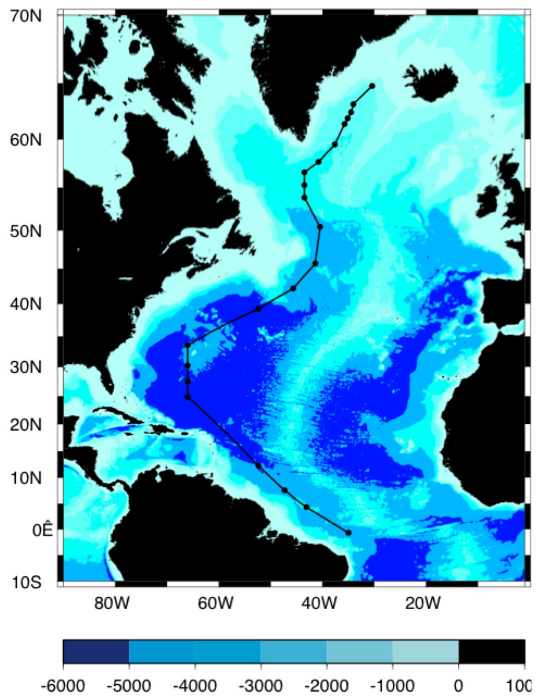
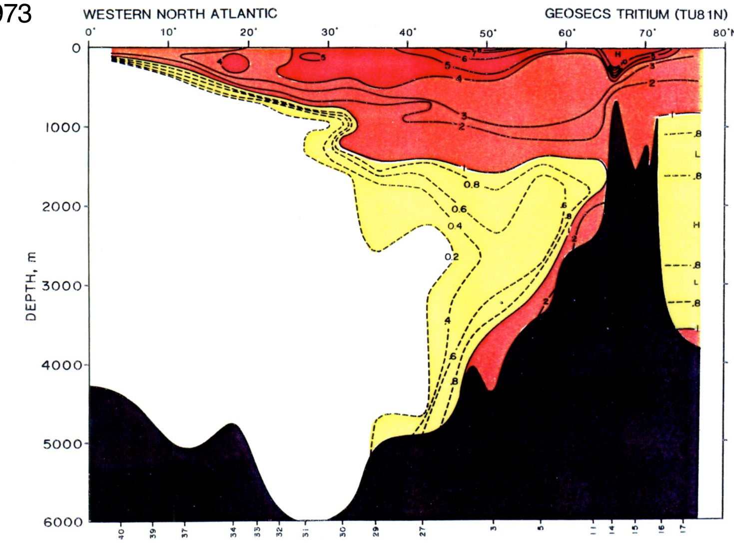


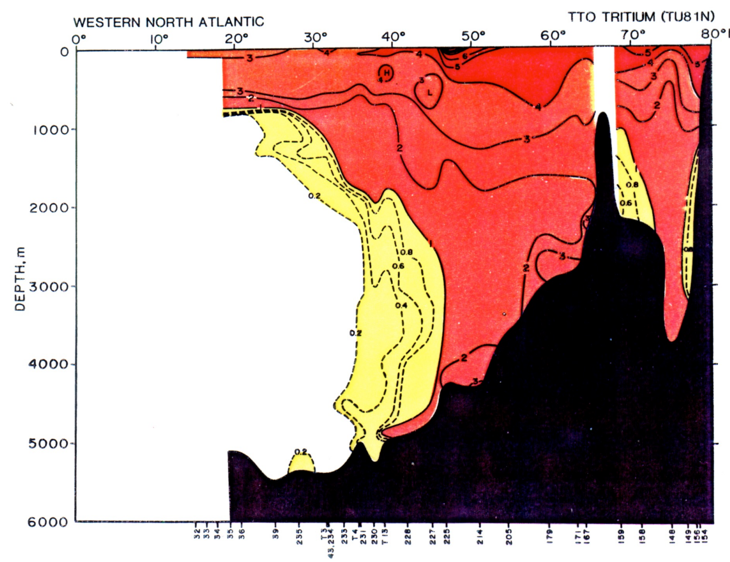
Fig. 4. Tritium and ^{90}Sr concentrations in North Atlantic surface water, 20°–60°N, over the period 1952–1974. Circles are observed values (Table 2, columns 4 and 5, estimated uncertainties ~±5%). Curves are obtained from the model, cf. text (numerical values listed in Table 2, columns 6 and 7). Dotted curve = ^{90}Sr/tritium ratio normalized to 1974 value. The curves are for model parameters $1/\kappa_1$ = 30 years, $1/\kappa_2$ = 2.5 years, $\gamma = 0.35$.



1972-1973



1981



$^3H/^3He$ dating

Tritium was produced during atmospheric bomb testing and entered the ocean as precipitation.

Tritium is radioactive with $t_{1/2}$ = 12.33 y (λ = 1.78 x 10^{-9} s^{-1})

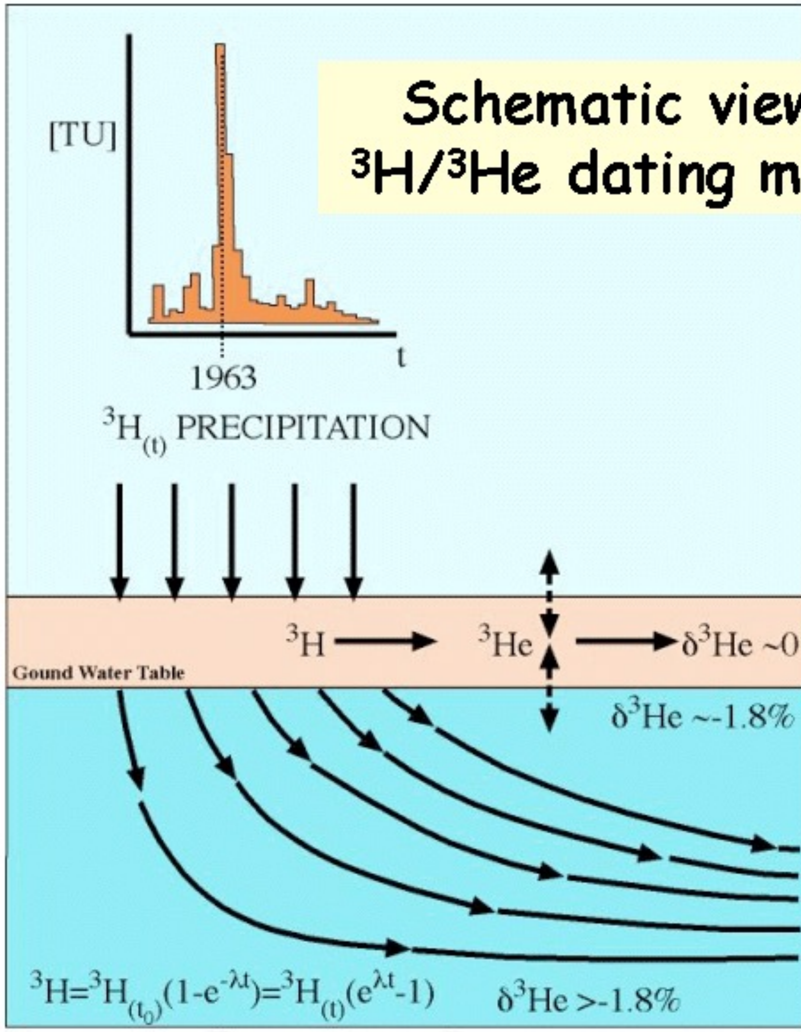
The daughter product ^3He is stable.

In contact with the atmosphere [^3He] is assumed to be in equilibrium with the atmosphere.

The age can be calculated as follows:

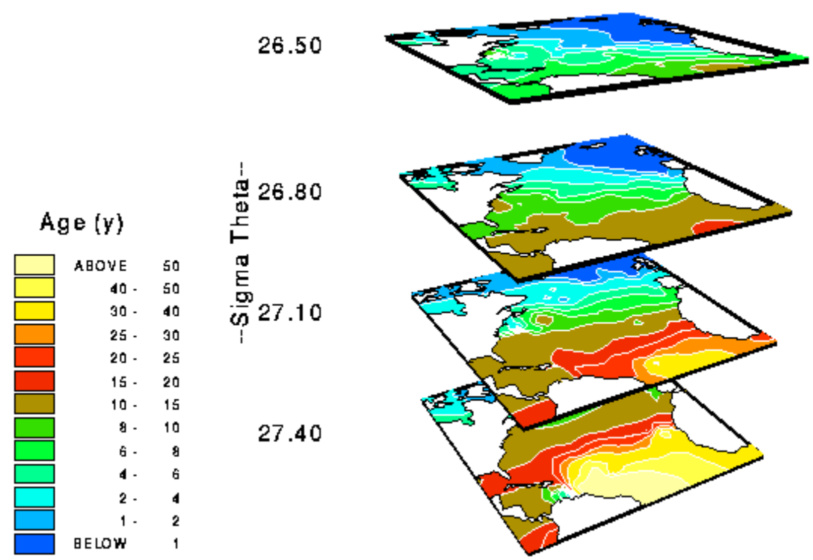
$$\boxed{\text{Age} = \frac{t_{1/2}}{\ln 2} \ln\left(1 + \frac{^3\text{He}}{^3\text{H}}\right)}$$

Schematic view of $^3H/^3He$ dating method

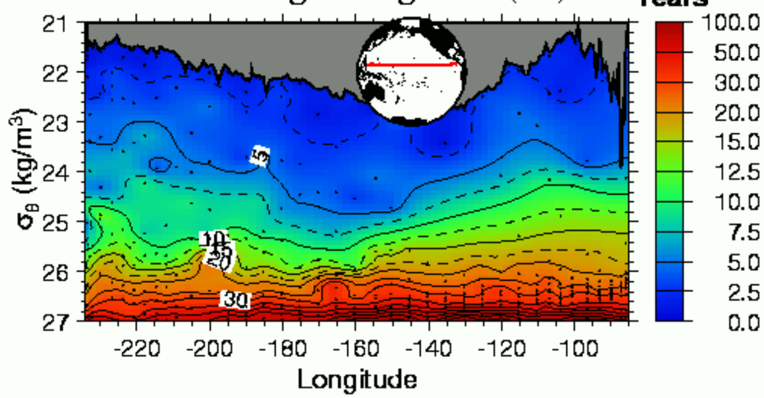


$$\tau = 1/\lambda \bullet \ln\left[1 + [^3He]/[^3H]\right] \qquad \lambda = T_{1/2}/\ln_2$$

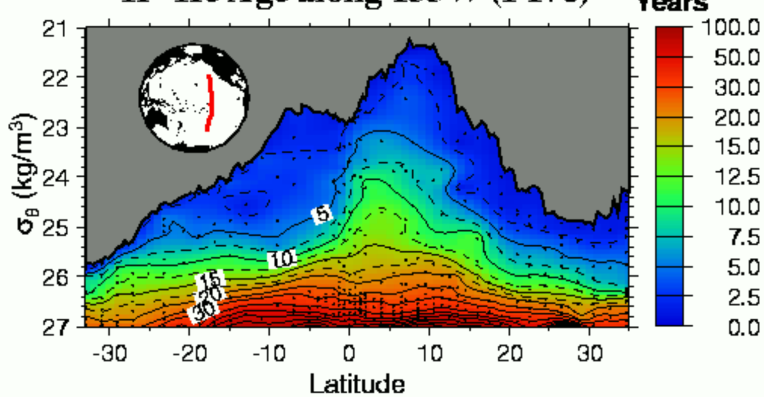
North Atlantic T-He Age (y)



^{3}H-^{3}He Age along 9.5N (P4)



^{3}H-^{3}He Age along 135W (P17c)



WHOI
Helium Isotope Laboratory

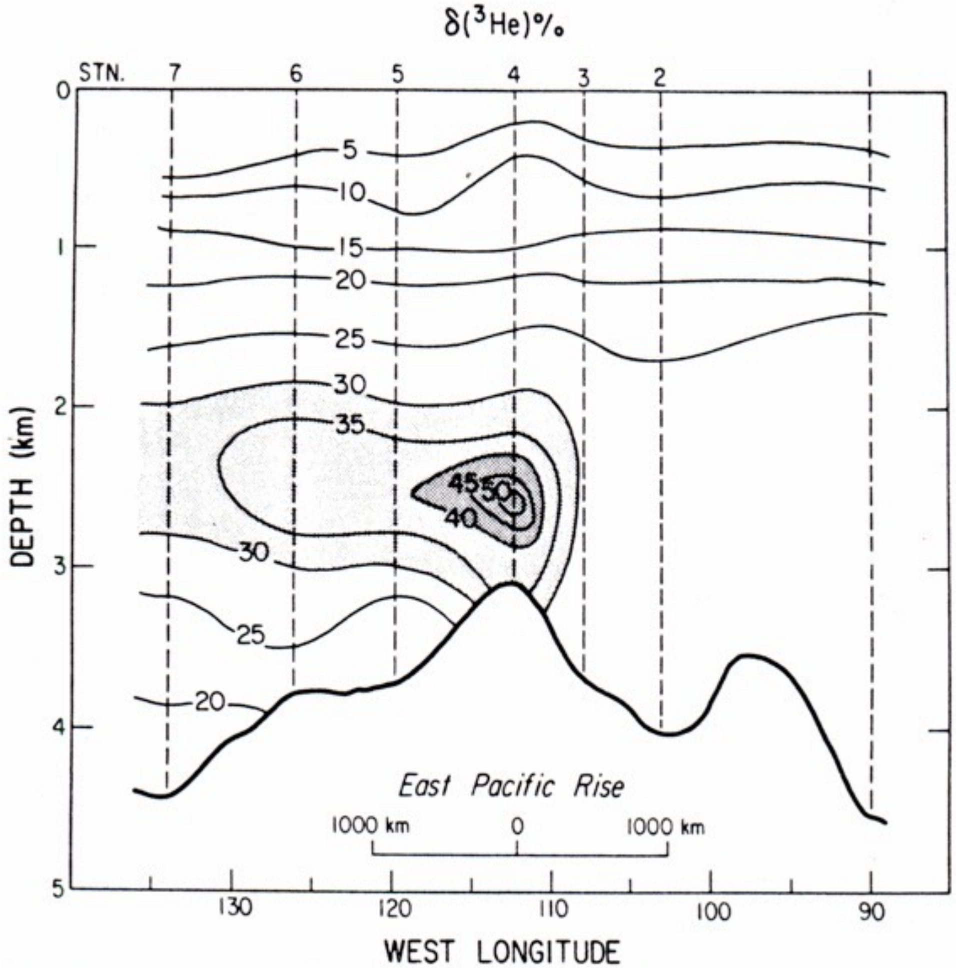


Fig. 2. Vertical section of the δ^3He distribution at 15°S across the East Pacific Rise.

Rates and Mechanisms of Water Mass Transformation in the Labrador Sea as Inferred from Tracer Observations*

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Lamont-Doherty Earth Observatory and Department of Earth and Environmental Sciences, Columbia University, Palisades, New York

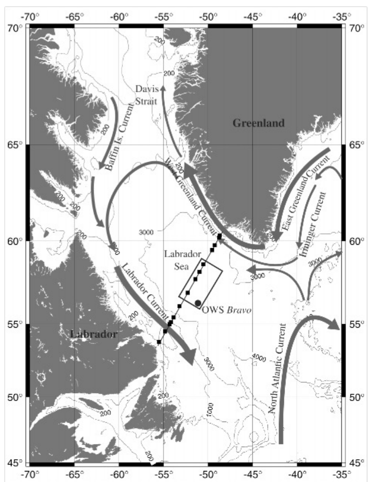


FIG. 1. Schematic of the circulation in the Labrador Sea. Major currents are indicated. Also shown is a typical WOCE AR7W cruise track and the nominal position of OWS Bravo. Solid squares show locations of stations occupied in Jun 1993 and at which samples for ^{3}H and ^{3}He were collected. Region used to define "central Labrador Sea" is marked by a rectangle.

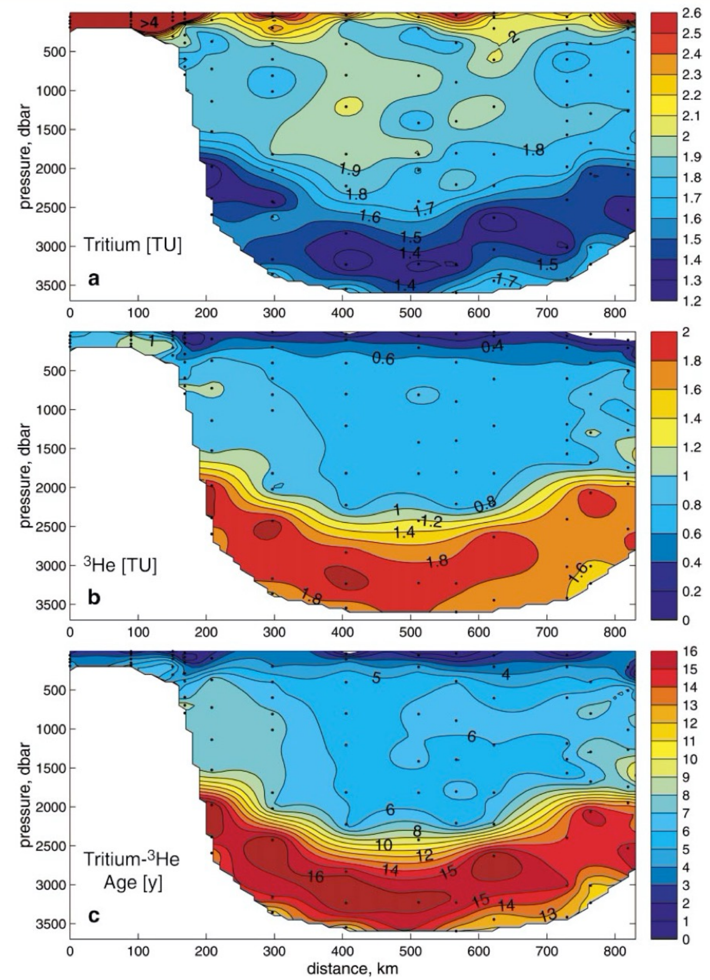
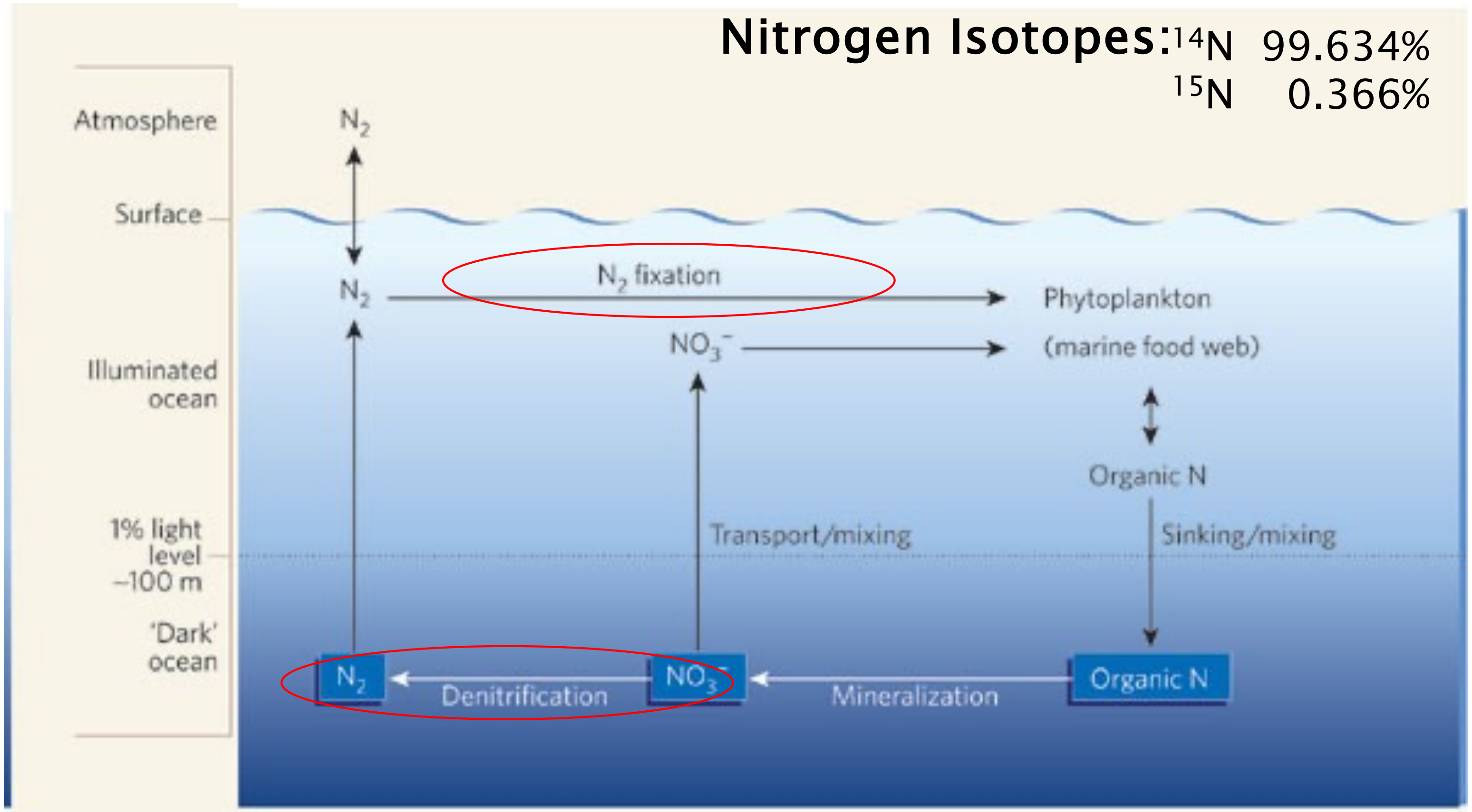


FIG. 4. Sections of (a) ^{3}H, (b) ^{3}He, and (c) ^{3}H–^{3}He age across the Labrador Sea along the WOCE AR7W section occupied in Jun 1993.

Schematic of Ocean Nitrogen Cycle

Nitrogen Isotopes: ^{14}N 99.634%
^{15}N 0.366%



The Global Nitrogen Budget–one example

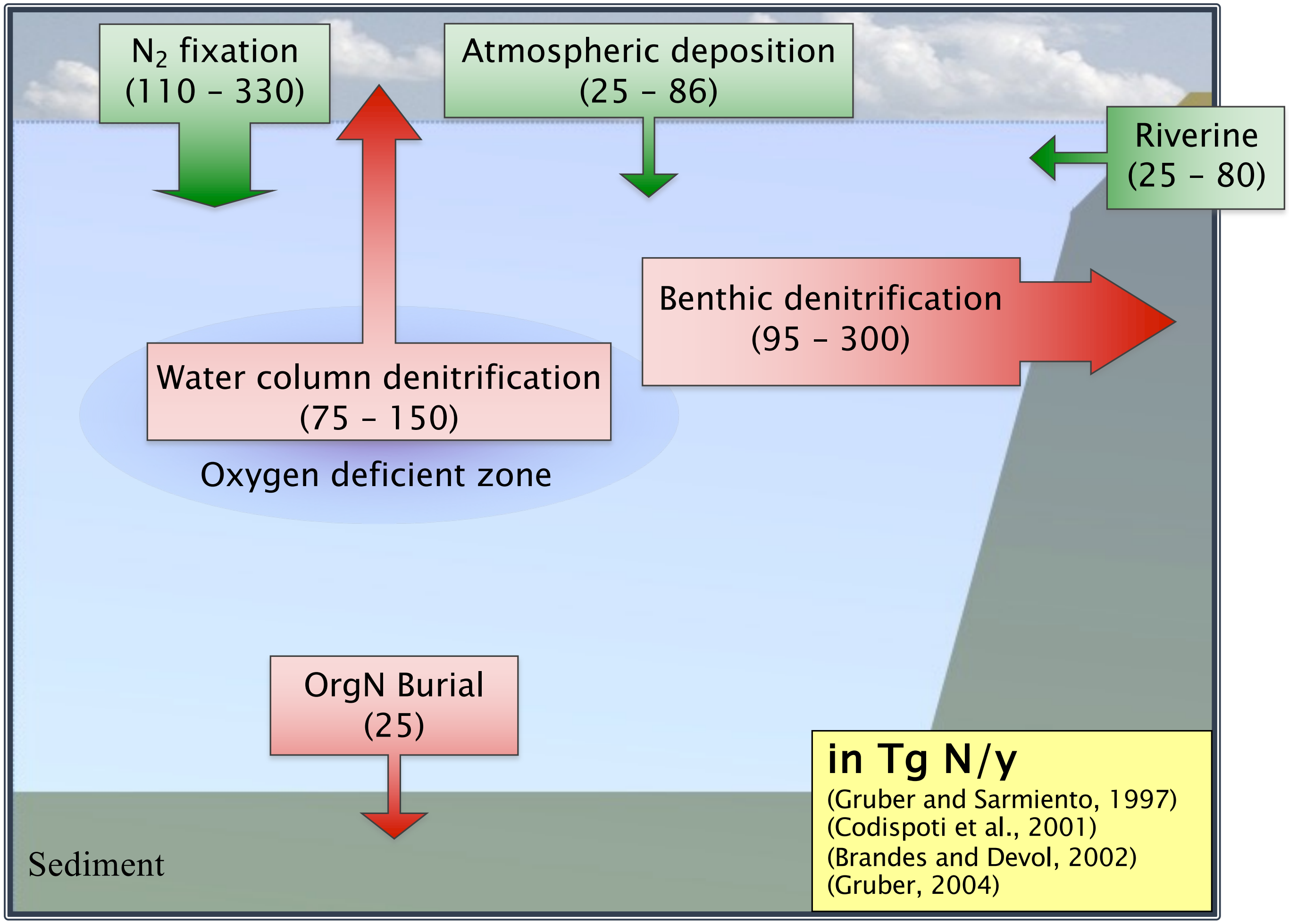
(Brandes et al, 2002)

Table 1. Fluxes and Isotopic Values for Source and Sink Nitrogen Budgets[a]

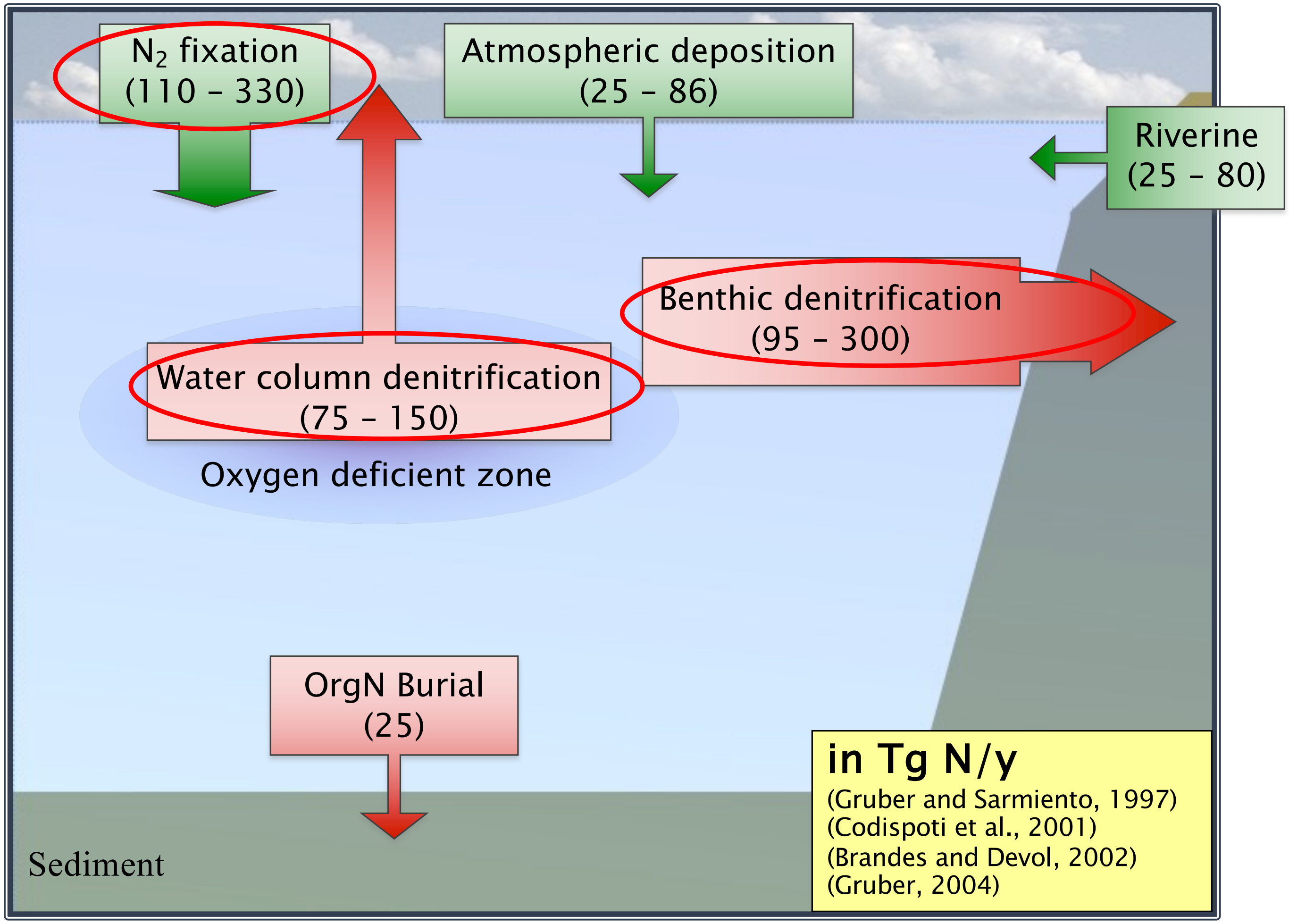
Term	Flux, Tg N yr^{-1}	Isotopic Value, ‰
Riverine source	25	4 (±4)
Atmosphere sources (DON + DIN)	25	−4 (±5)
Nitrogen fixation	110–330	−1 (±1)
Total sources	160–380	−1
Sedimentary denitrification	−200–280	3.5 (±2)
Water column denitrification	−75	−20 (±3)
Organic burial	−25	6 (±4)
Total sinks	−300–380	−1
Net	−200–0	−1 (±2)

[a]Values in parentheses for isotopic values are estimated variances for each term. See text for references.

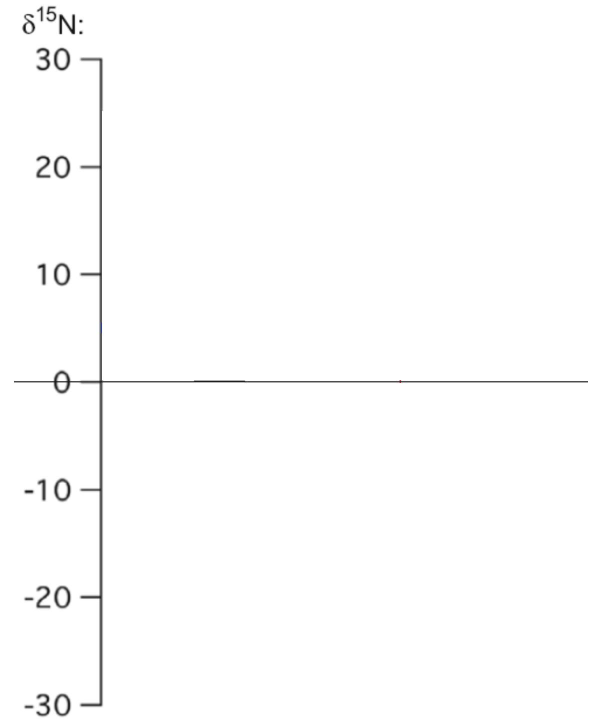
N Budget: Sources and Sinks



N Budget: Sources and Sinks

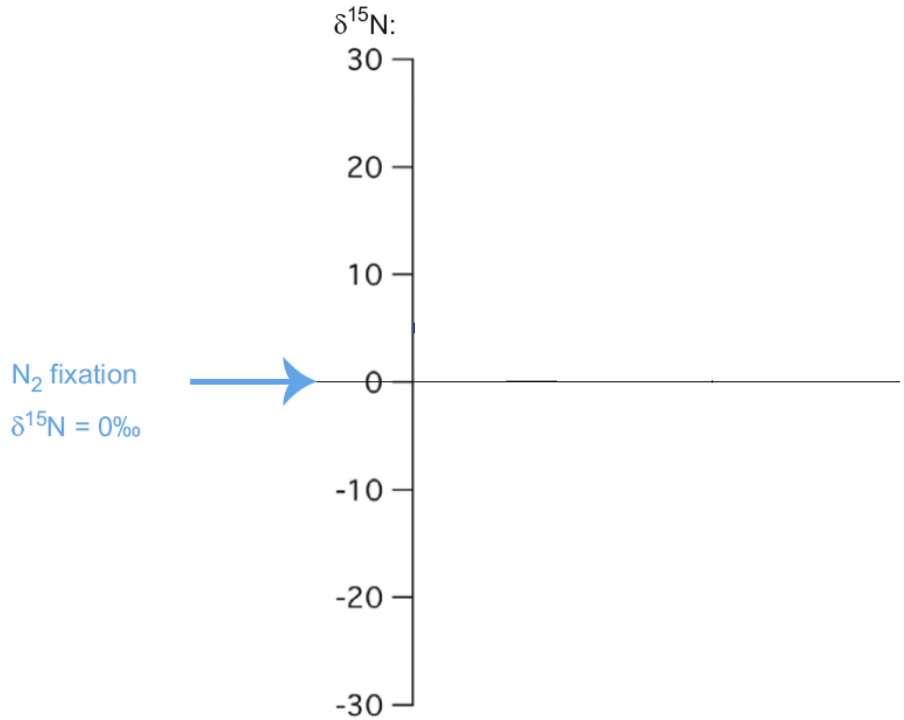


Marine N isotope budget



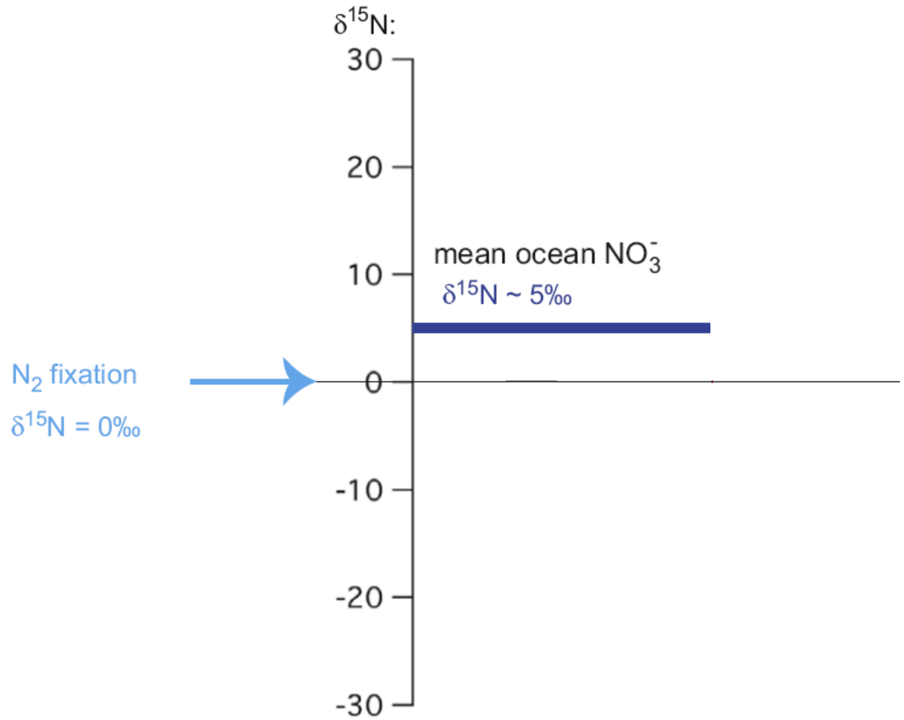
(Brandes and Devol 2002)
(Sigman et al. 2009)

Marine N isotope budget



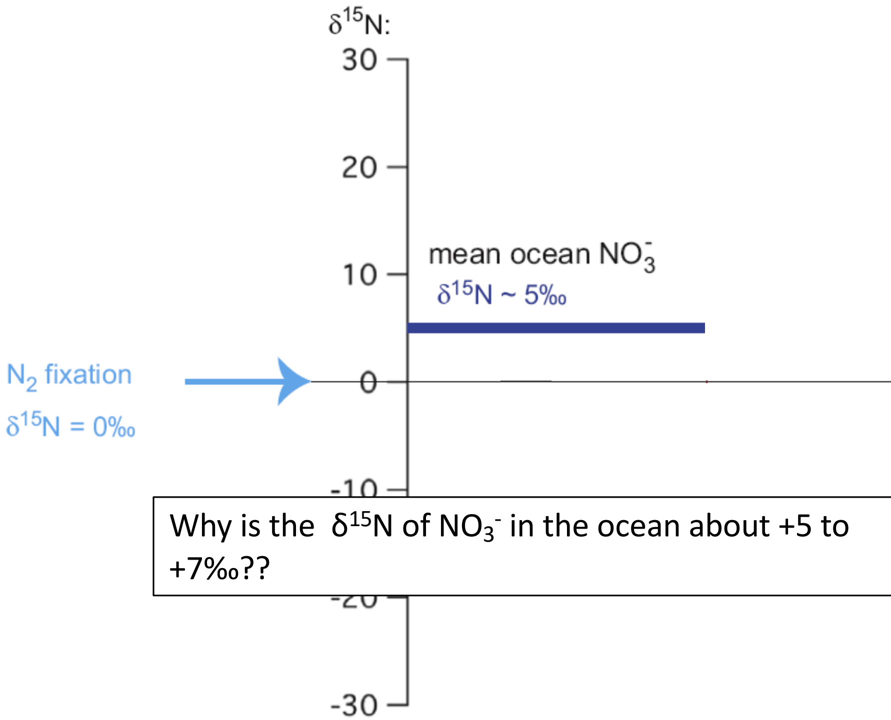
(Brandes and Devol 2002)
(Sigman et al. 2009)

Marine N isotope budget



(Brandes and Devol 2002)
(Sigman et al. 2009)

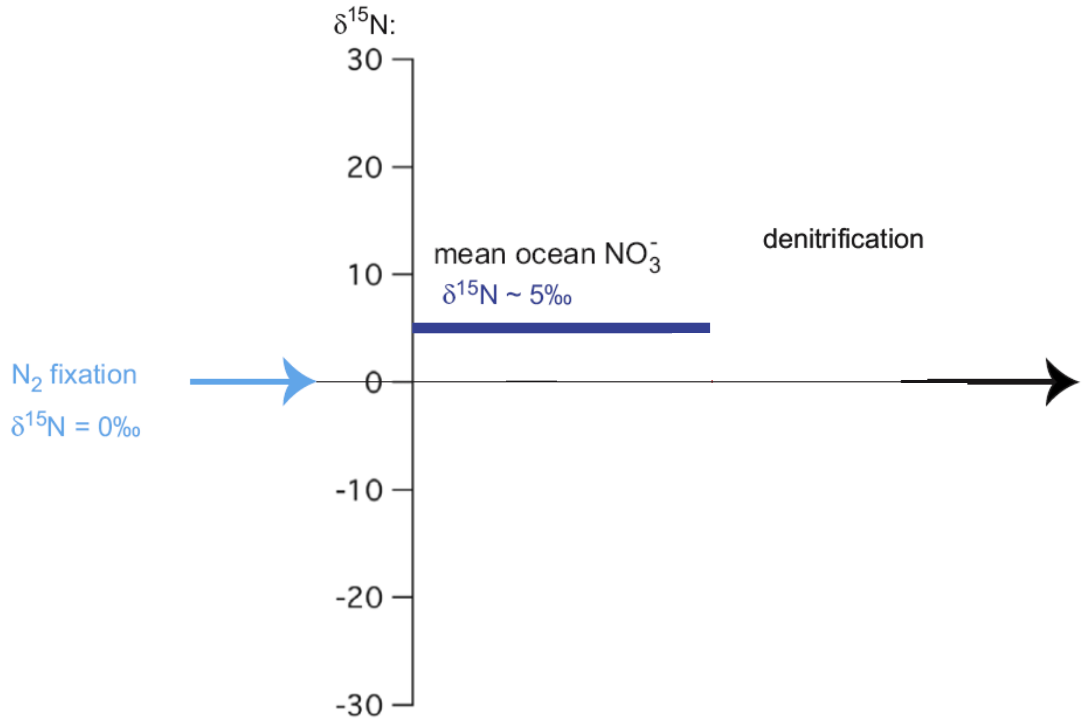
Marine N isotope budget



Why is the $\delta^{15}N$ of NO_3^- in the ocean about +5 to +7‰??

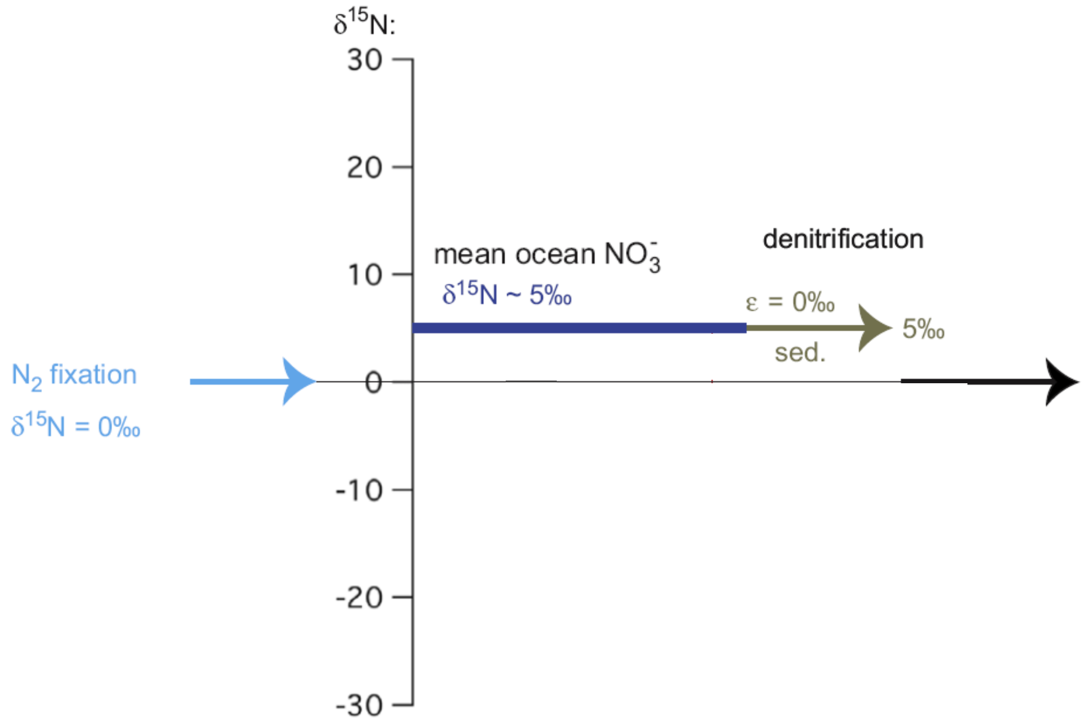
(Brandes and Devol 2002)
(Sigman et al. 2009)

Marine N isotope budget



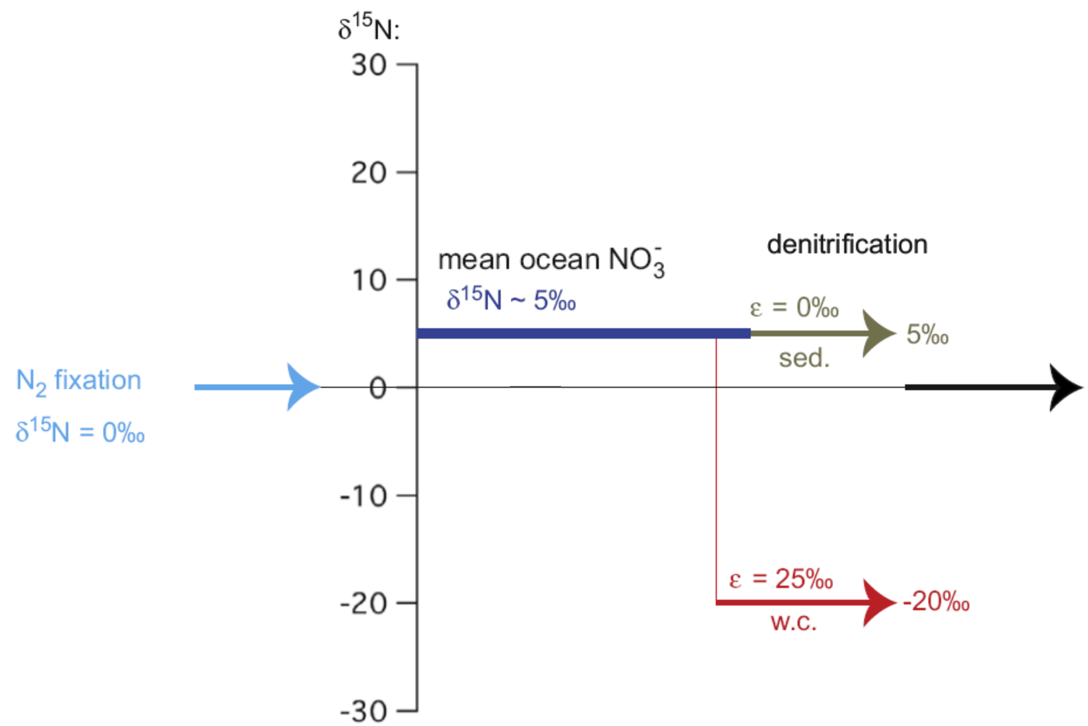
(Brandes and Devol 2002)
(Sigman et al. 2009)

Marine N isotope budget



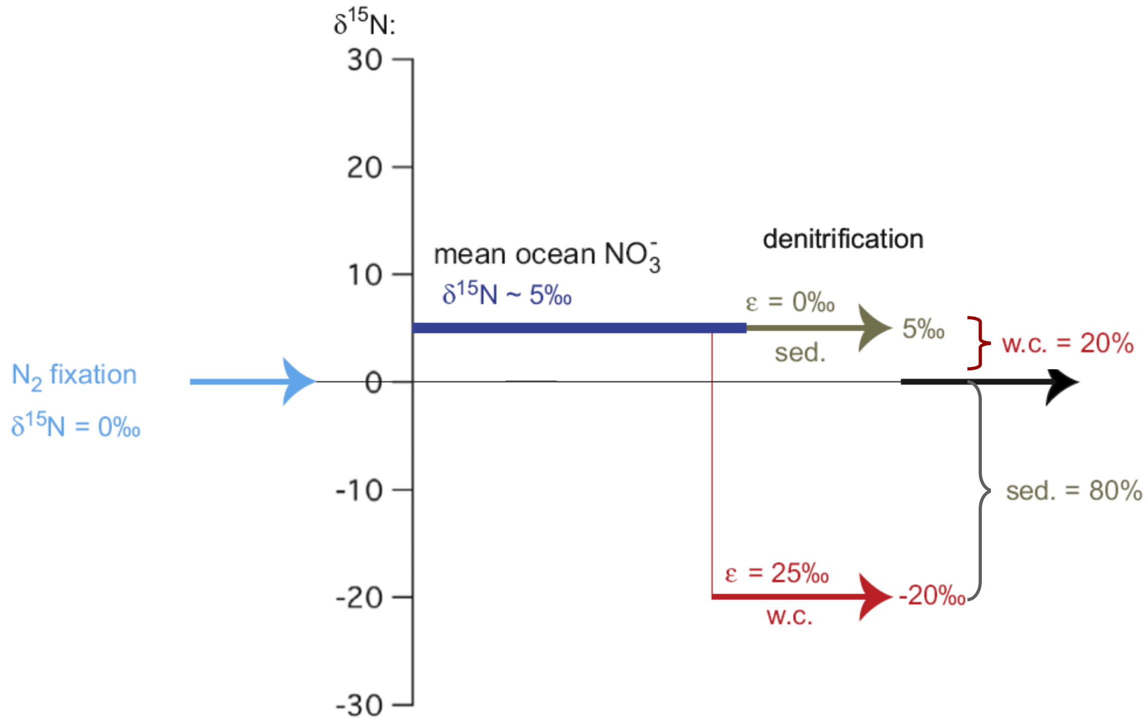
(Brandes and Devol 2002)
(Sigman et al. 2009)

Marine N isotope budget



(Brandes and Devol 2002)
(Sigman et al. 2009)

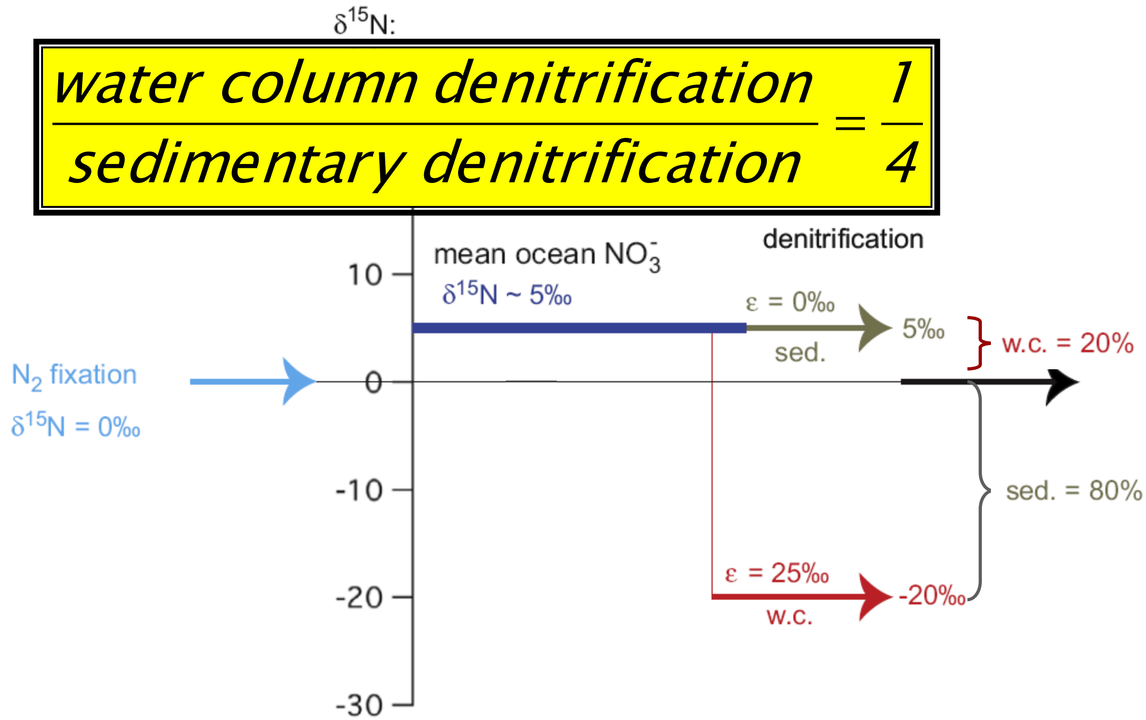
Marine N isotope budget



(Brandes and Devol 2002)
(Sigman et al. 2009)

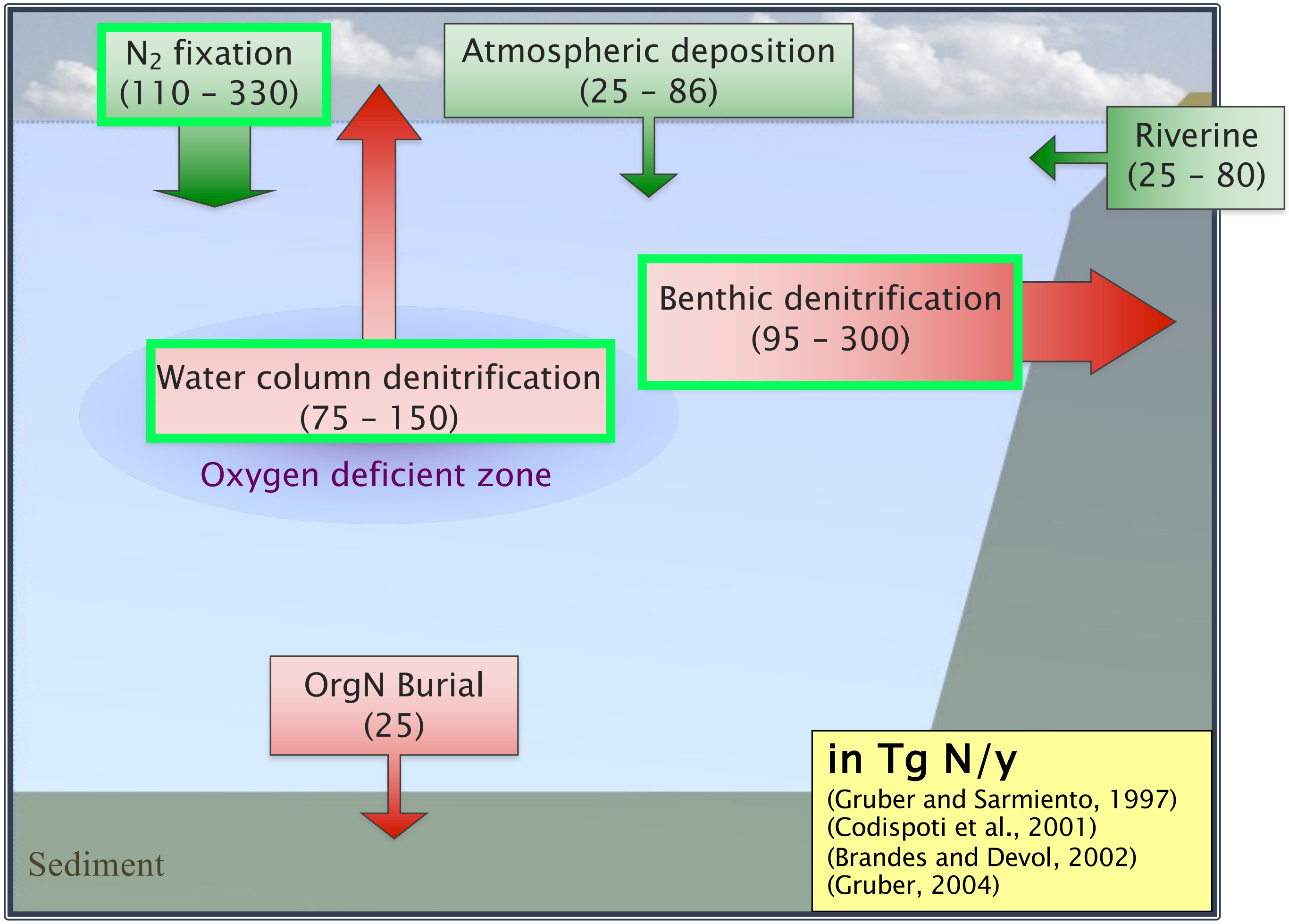
Marine N isotope budget

$$\frac{\textit{water column denitrification}}{\textit{sedimentary denitrification}} = \frac{1}{4}$$

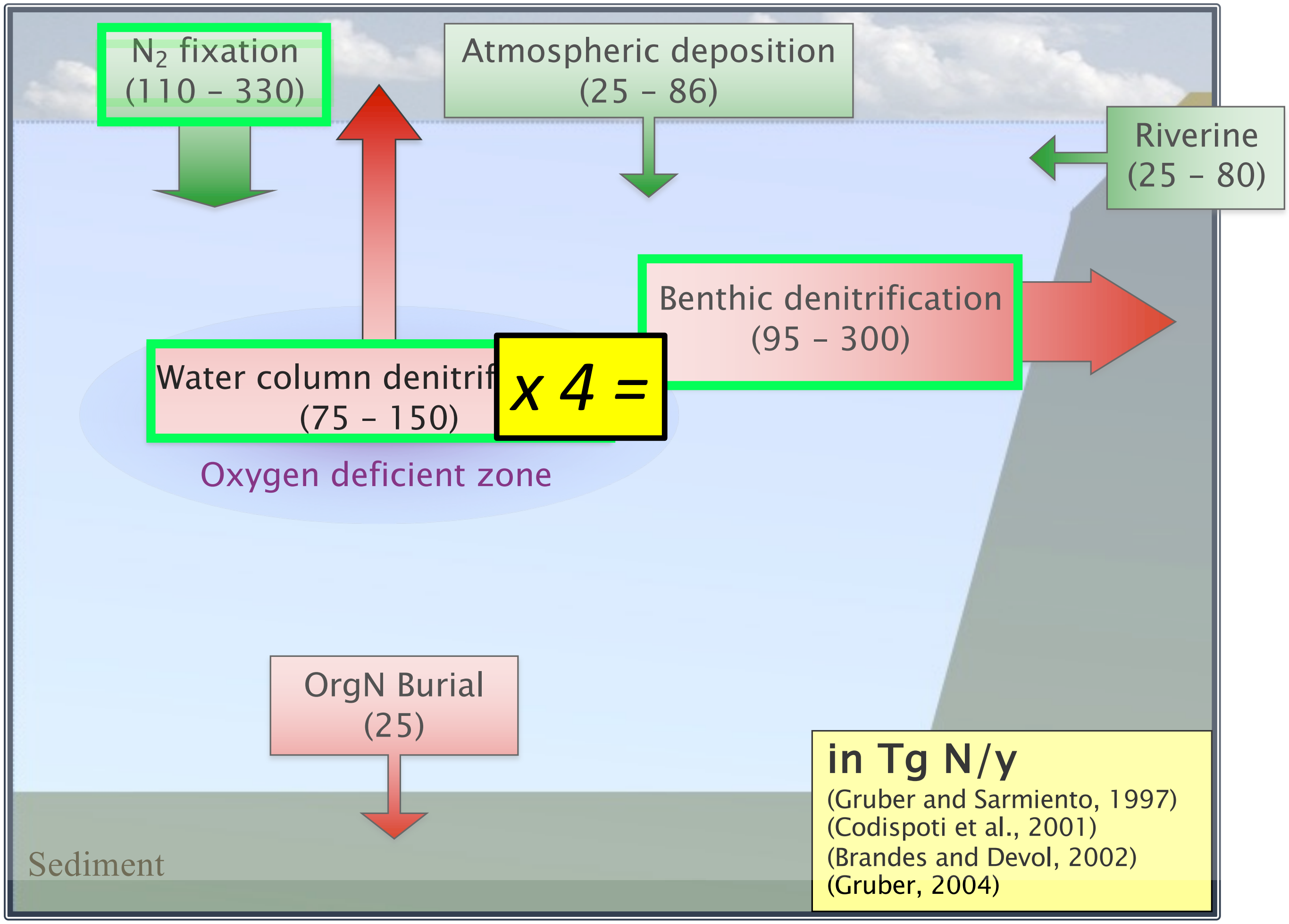


(Brandes and Devol 2002)
(Sigman et al. 2009)

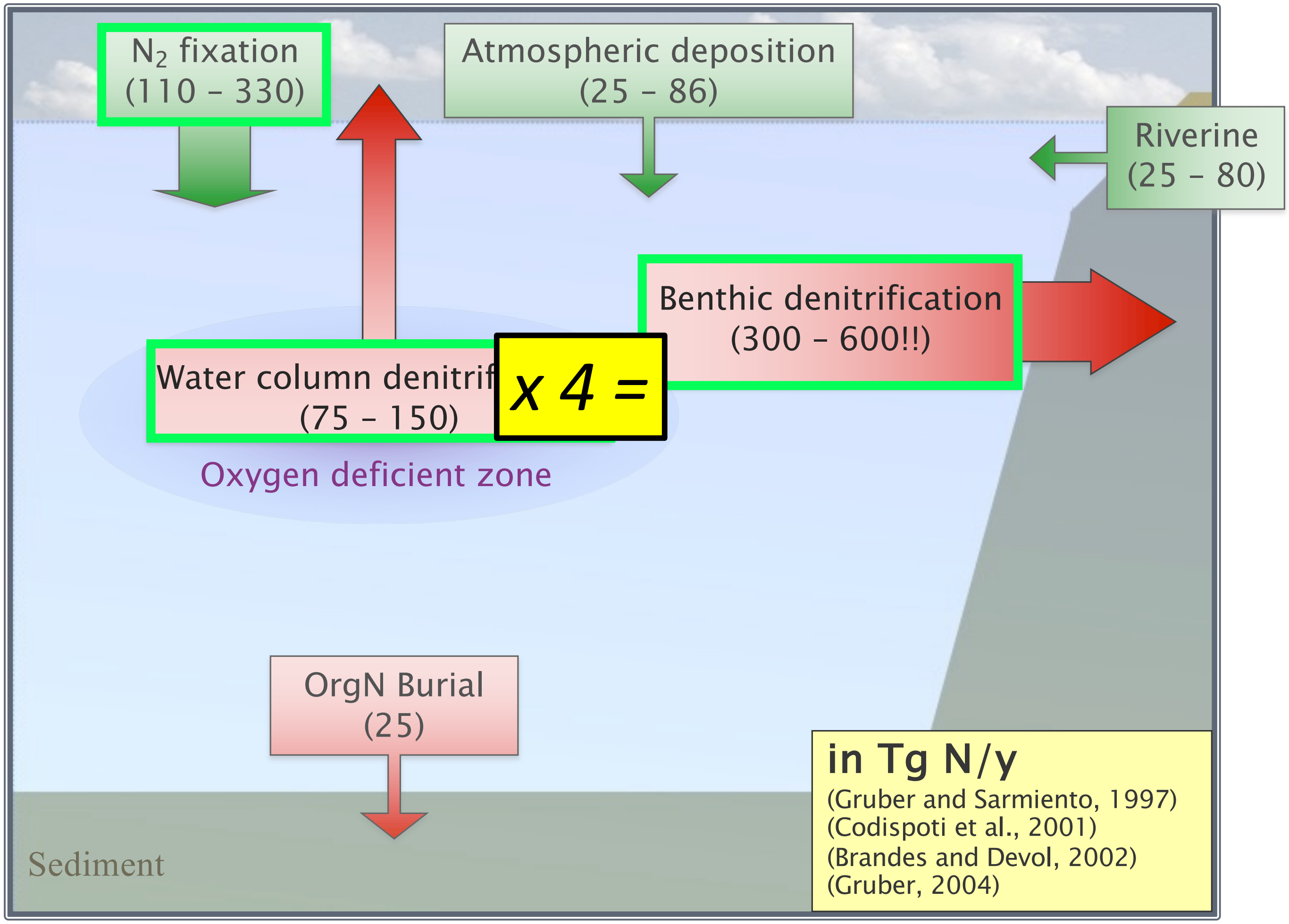
The Marine Nitrogen Budget



The Marine Nitrogen Budget



The Marine Nitrogen Budget



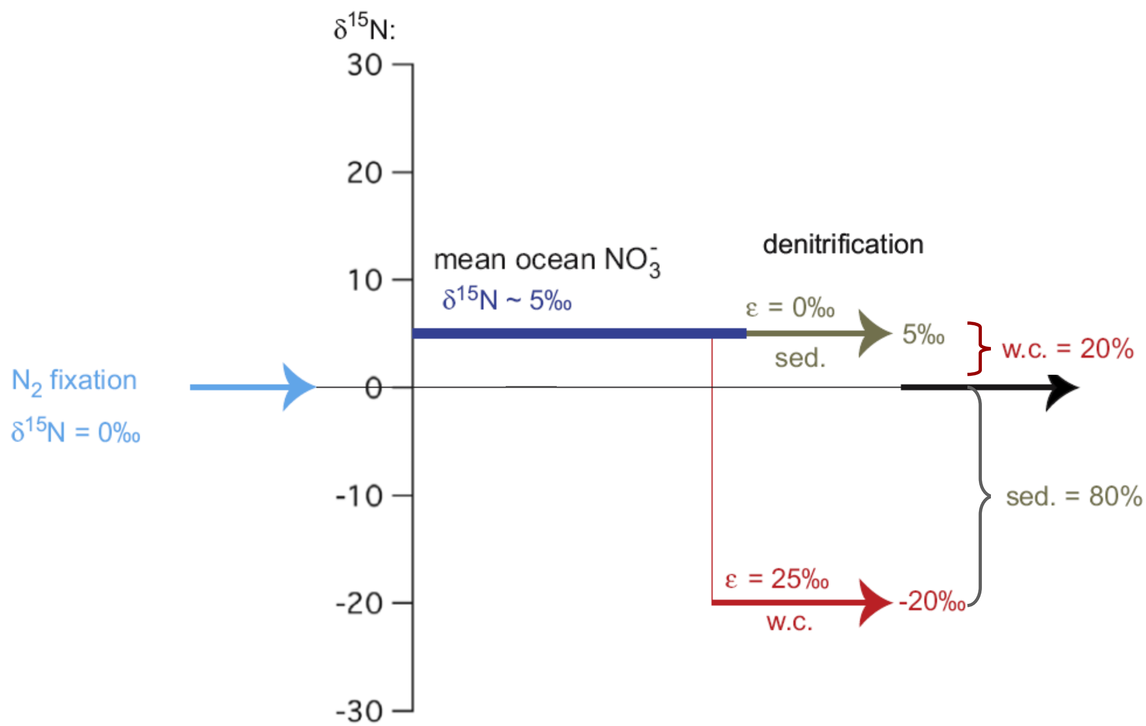
The Marine Nitrogen Budget

Water column denitrification	Benthic denitrification	Net (Tg N/y)	Time to deplete N inventory
75	300	> −200	< 3000 y

The Marine Nitrogen Budget

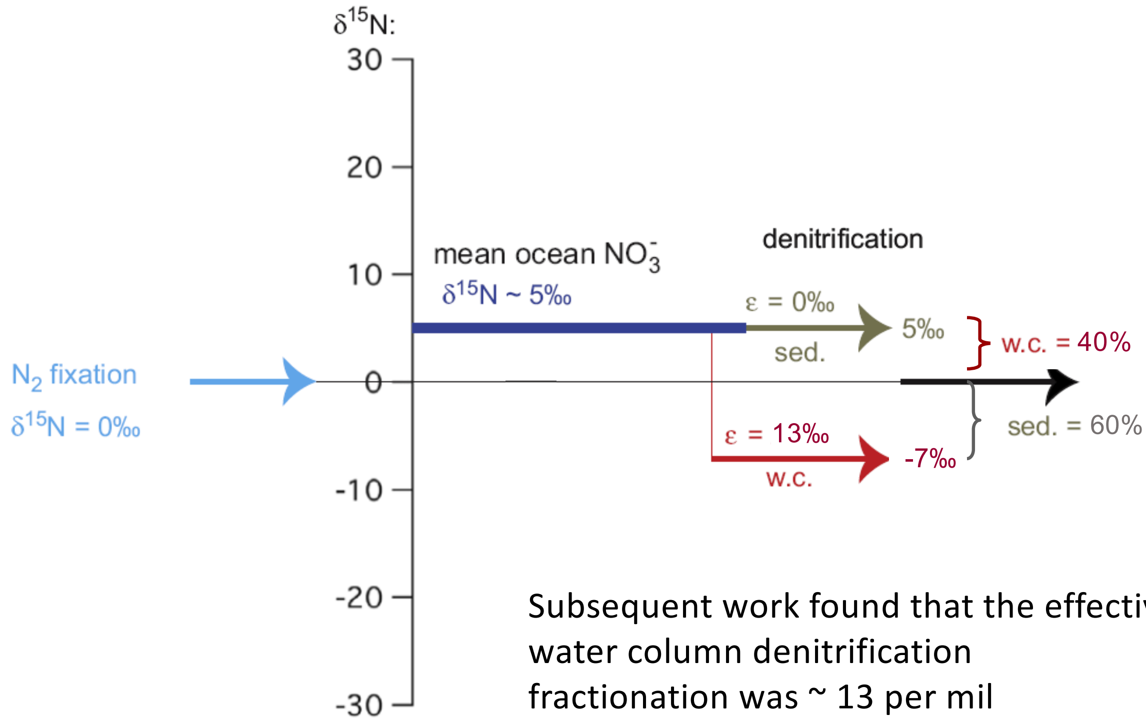
Water column denitrification	Benthic denitrification	Net (Tg N/y)	Time to deplete N inventory
75	300	> −200	< 3000 y
150	600	> −575	< 1000 y

Marine N isotope budget



(Brandes and Devol 2002)
(Sigman et al. 2009)

Marine N isotope budget



Subsequent work found that the effective water column denitrification fractionation was ~ 13 per mil

(Brandes and Devol 2002)
(Sigman et al. 2009)

The Marine Nitrogen Budget

Water column denitrification	$\varepsilon_{W.C.}$	W.C.:Benthic	Benthic denitrification	Net (Tg N/y)
75	25‰	1:4	300	~ −200

With a water column denitrification fractionation of ~ 13 per mil

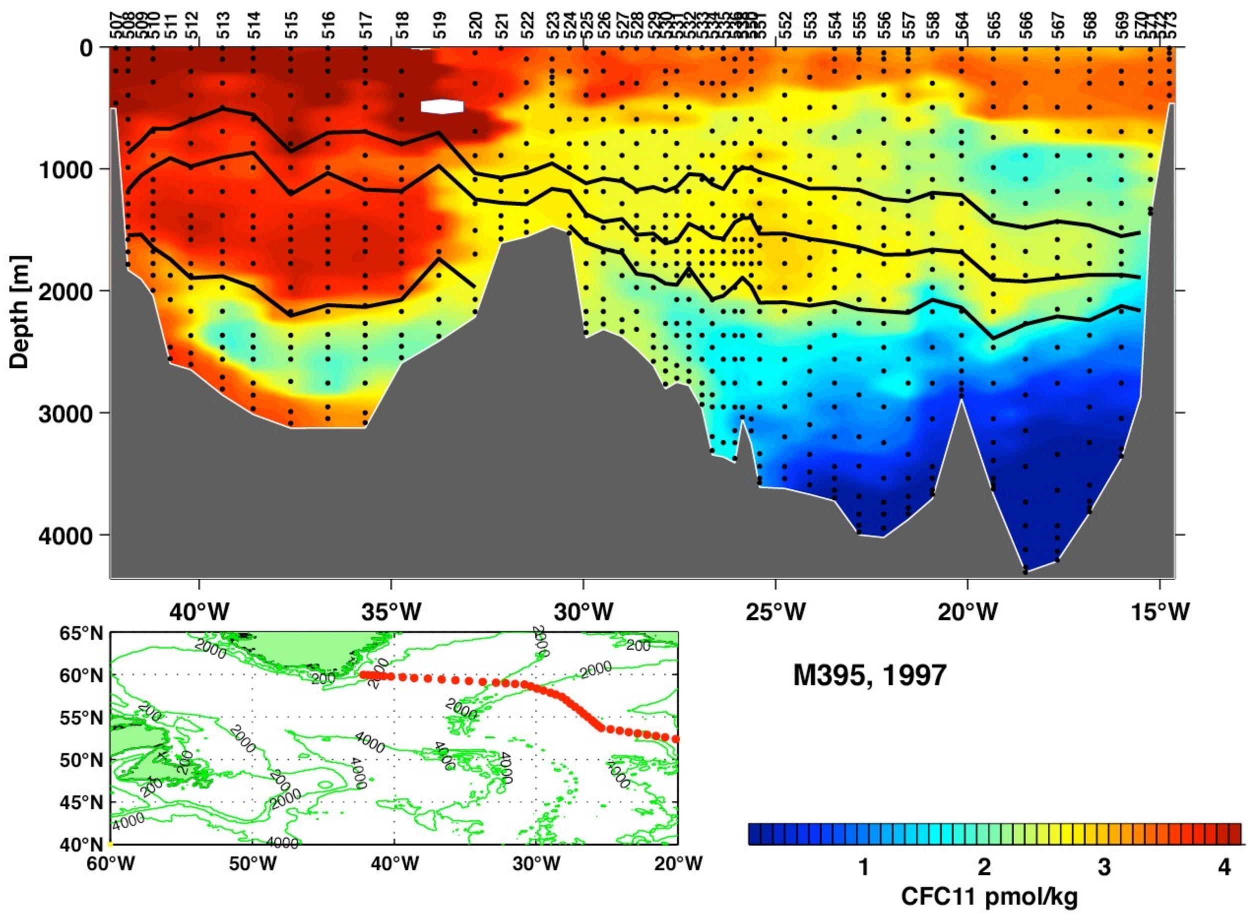
The Marine Nitrogen Budget

Water column denitrificatio n	$\varepsilon_{W.C.}$	W.C.:Benthic	Benthic denitrification	Net (Tg N/y)
75	25‰	1:4	300	~ −200
75	13‰	1:1.5	113	~ −13

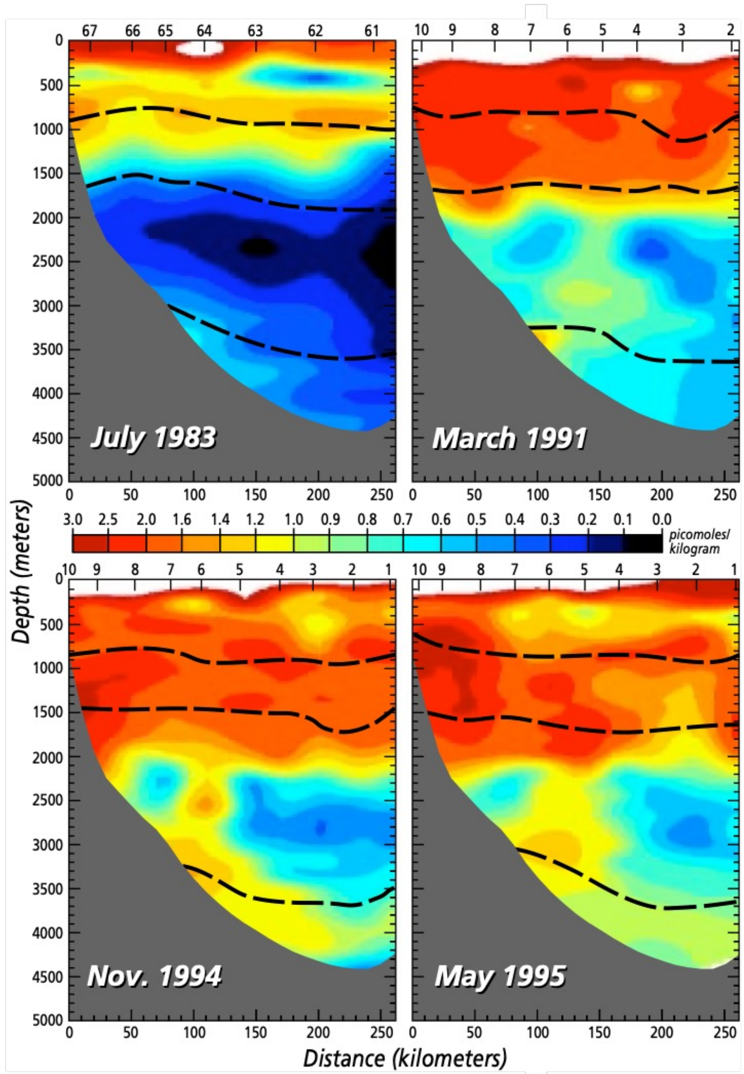
This closes the Nitrogen budget!

Altabet et al. 2007

Pathway of Labrador Sea Water



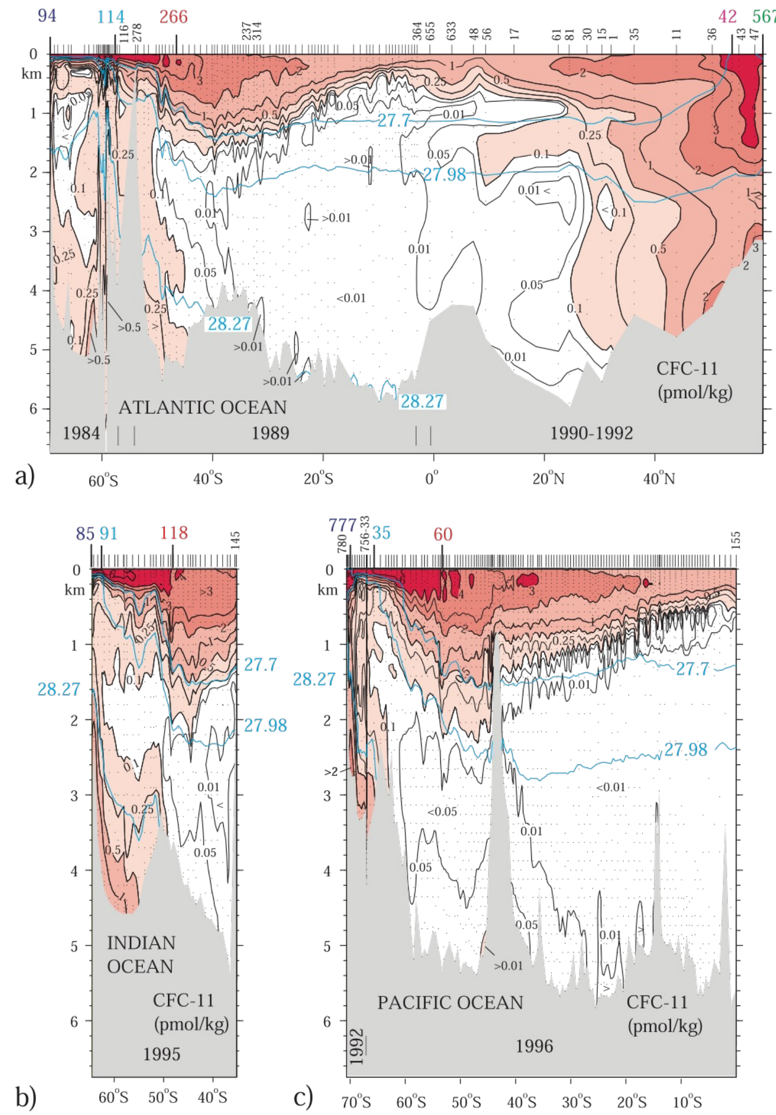
Rhein et al. (2002)



Four chlorofluorocarbon (CFC) sections taken at various times along 55°W south of the Grand Banks. Note the absence of any significant CFC signal at the depth of the Labrador Sea Water (about 1,500 meters depth) in 1983, but the sudden flooding of these depths with CFCs in the later sections, as newly formed Labrador Sea Water flows around the Grand Banks and into the Sargasso Sea. These changes correspond to the tritium increases seen in the Bermuda time series (see figure opposite).

Jenkins and Smethie, Oceanus, 1996

Vertical distributions of CFC-11 (pmol kg^{-1}) measured along hydrographic lines in the (a) Atlantic, (b) Indian, and (c) Pacific oceans.



Orsi et al., 2002

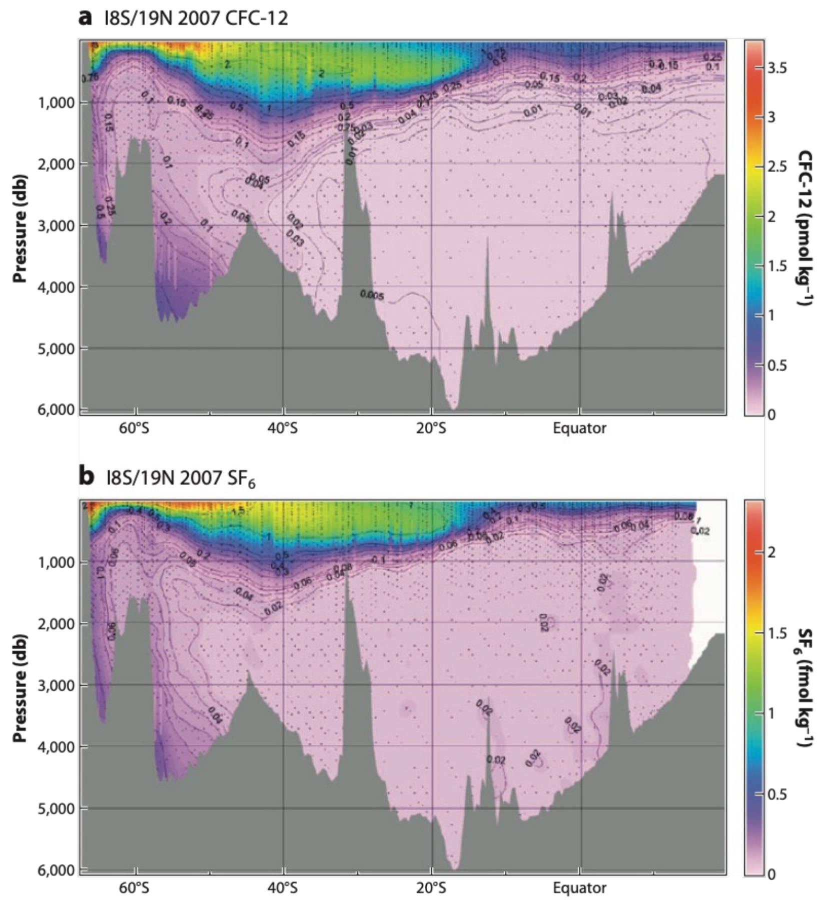


Figure 2

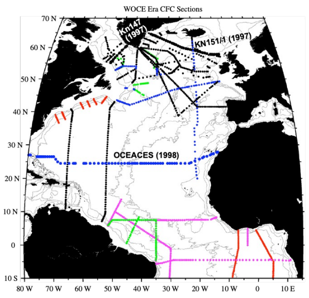
(*a*) Section of dissolved CFC-12 along 90°E (the I8 S/I9N hydrographic sections) in the South Indian Ocean collected February–April 2007. (*b*) Dissolved SF_6 along the same section shown in (*a*). Black dots indicate sampling stations. Figure adapted from Bullister & Wisegarver (2008).

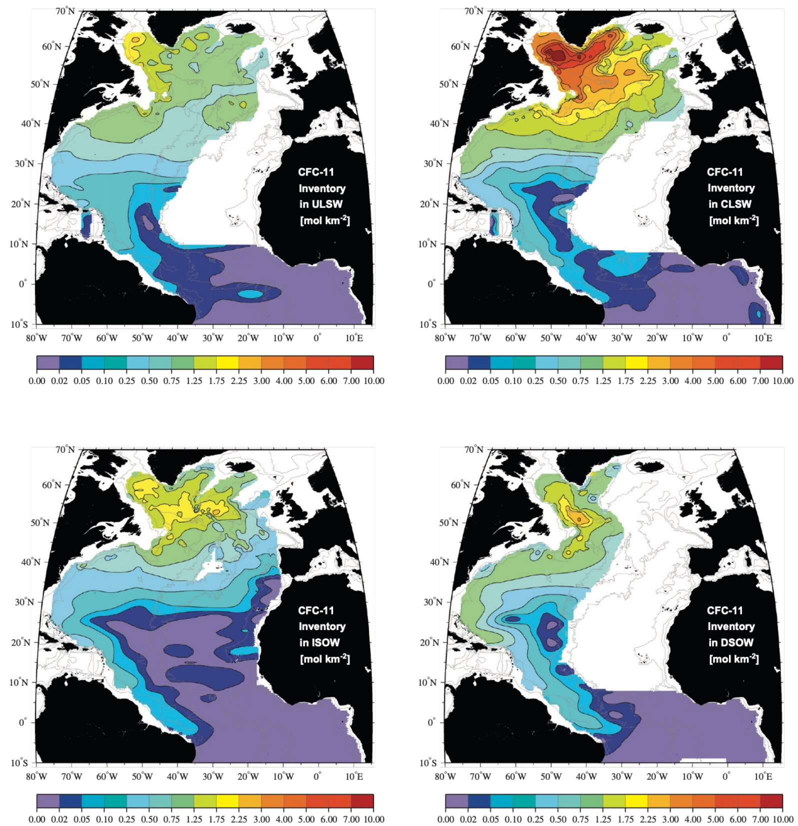
Fine, 2010

Formation rate of NADW

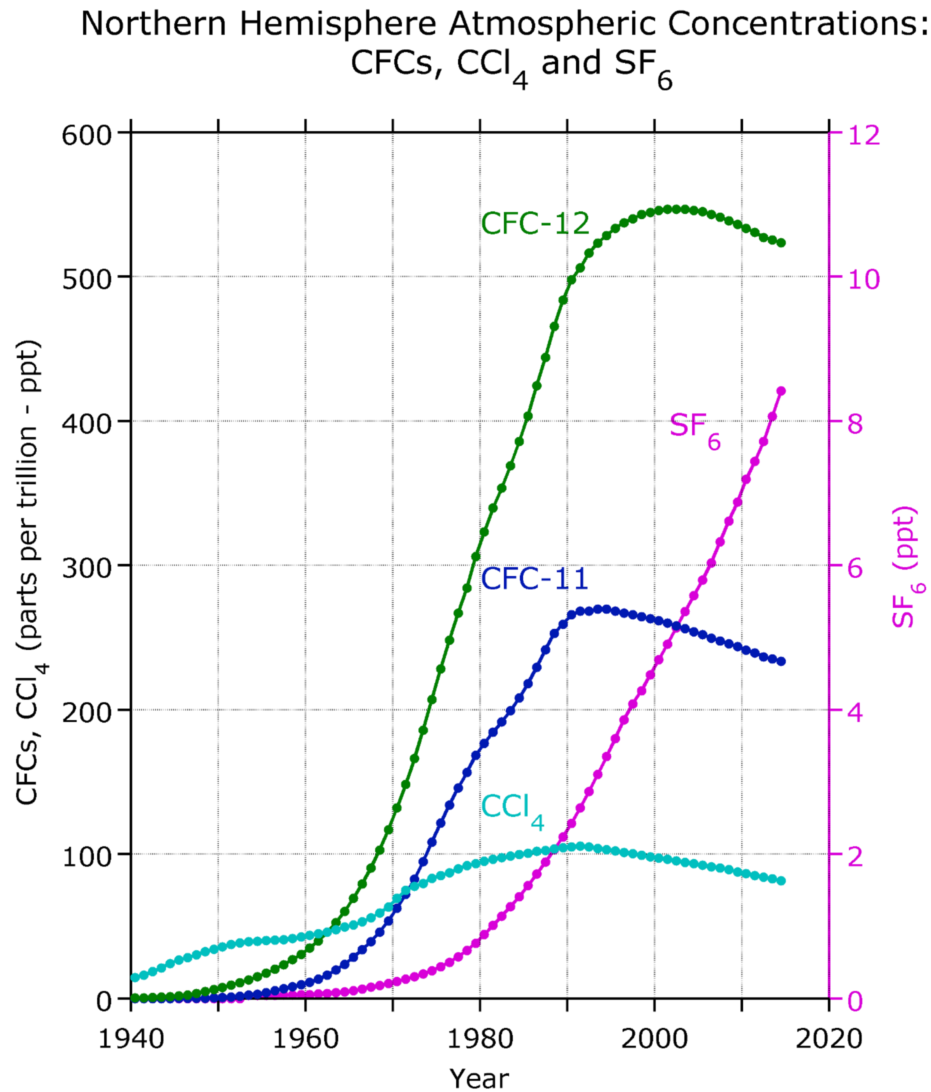
$$I = \int R(t)C(\theta, S, t)dt$$

I: Water mass CFC inventory
R: Rate of water mass formation
C: Source water CFC concentration
θ: Potential temperature
S: Salinity
t: Time





LeBel et al., 2008



- CFCs – Cholorfluorocarbons (refrigerant)
- CCl_4 – Carbon Tetracholoride (fumigant)
- SF_6 – Sulphur hexafluoride (semi-conductors)

Water Mass	CFC-11 Inventory (million mol) 1970-1990 (Smethie and Fine, 2001)	CFC-11 Inventory (million mol) 1970-1997 (LeBel et al., 2008)	Formation Rate 1970-1990 (Sv) (Smethie and Fine, 2001)	Formation Rate 1970-1997 (Sv) (LeBel et al., 2008)
Upper LSW	4.2	10.5	2.2	3.5
Classical LSW	14.7	23.4	7.4	8.2
ISOW	5	10.4	5.2	5.7
DSOW	5.9	8.3	2.4	2.2
NADW (total)	29.8	52.6	17.2	19.6

LSW - Labrador Sea Water
ISOW - Iceland Scotland Overflow Water
DSOW - Denmark Strait Overflow Water